BASELINE PROJECT PLAN

Smart Recruitment and Vacancy Management System with Automatic Job Matching

INSY281: System Analysis & Design | Lecturer: Mr. Jefferson Mwatati | Report prepared October 2026

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Client / Sponsor	Managing Director, (TalentBridge Recruitment Ltd, Nairobi)
System type	Web-based Information System combining a Transaction Processing System (vacancies, applications, interviews), a Management Information System (reports) and a Decision Support System (automatic job matching).
Development approach	Agile (Scrum), delivered in eight 2-week sprints.
Budget and duration	Ksh 2,084,500 (including 10% contingency); 16 weeks, 11 January 2027 to 30 April 2027.
GitHub repository	https://github.com/victorious23-lucky/SmartRecruitmentBPP.git

1. Introduction

1.1 Brief Overview and Purpose

TalentBridge Recruitment Ltd connects employers with job seekers. It handles about 150 vacancies and receives about 2,000 applications every month, mostly by email with CVs attached and tracked in Excel sheets. This project will build a centralized web platform where employers post vacancies, job seekers create profiles and apply online, and recruiters manage every application through a clear pipeline. An automatic matching engine reads CVs, compares candidate skills, experience, education and location with vacancy requirements, and ranks candidates with an explained score. The recruiter always makes the final decision.

1.2 Problem and Opportunity

- **Problem:** recruiters spend about 25 minutes screening each CV by hand, shortlisting a vacancy takes about 3 days, CVs are lost or duplicated in email, vacancy status is unclear, and good candidates in the old database are never reconsidered
- **Opportunity:** automated CV parsing and ranked matching can cut shortlisting time by at least 60%, reuse about 8,000 existing candidate records, and give management accurate hiring statistics.

1.3 Recommended Course of Action

Build a custom web system (MongoDB, Express.js, React, Node.js, plus a Python matching service) in 16 weeks, delivered in incremental sprints with client sign-off at each milestone

1.4 Project Scope

Scope element	Definition
Units affected	Recruitment department (12 recruiters); client employers' HR contacts (vacancy owners); job seekers (applicants); Management (reports); Administration (user and data management).
Interaction with other systems	Email and SMS gateway for notifications; hosted MongoDB and file storage for CVs; AI/NLP service for text understanding, called from the server only; one-time import of the existing Excel candidate database (about 8,000 records); calendar (.ics) files for interview invitations.
Range of capabilities (in scope)	4 roles (Administrator, Recruiter, Employer, Job seeker); vacancy creation, approval, publishing, expiry and closing; candidate profiles and CV upload (PDF/DOCX); application pipeline; automatic matching with explanations; interview scheduling; notifications; dashboards; audit trail.
Out of scope	Automatic hiring or rejection decisions, video interviewing, payroll and invoicing, native mobile apps, payment processing.

Scope / Major Project Activities

Activity	Description
Planning and requirements	Interviews with recruiters, employers and applicants; review of the current process; feasibility study; requirements specification.
System design	Use-case, activity, sequence and class diagrams; data-flow diagrams; MongoDB schema (9 collections); wireframes; REST API specification; matching-score design.
Core development	Authentication and roles, vacancy management, candidate profiles, CV upload and parsing, application pipeline.
Matching and reporting	Matching engine with explanations, interview scheduling, notifications, dashboards and reports.
Testing and evaluation	Functional, security, usability and performance tests; comparison of system shortlists with recruiters' manual shortlists.

Deployment and handover	Data migration, production deployment, training (3 sessions), user manual, final report and presentation.

2. System Description

2.1 Alternative Solutions

Option	Description	Assessment
A. Keep current process	Email, phone and Excel sheets.	No cost now, but slow, error-prone and impossible to measure. Rejected.
B. Subscribe to an international recruitment platform	Ready-made applicant tracking software paid per user per month.	Quick to start, but recurring foreign-currency fees, little local customization (Kenyan job categories, SMS) and less control over candidate data. Rejected.
C. Customize an open-source tracker	Adapt a free applicant tracking tool.	Low licence cost, but it lacks automatic matching and needs heavy changes plus server skills. Considered.

2.2 Narrative Description of the Proposed System

An employer or recruiter creates a vacancy with required skills, minimum experience, education level, location and closing date; a recruiter approves it and it is published on the careers portal. A job seeker registers, uploads a CV and completes a profile. The system extracts skills, experience and education from the CV into a structured profile, which the applicant can correct. When someone applies, or when a recruiter requests matches, the matching engine scores each candidate from 0 to 100 using weighted factors: skills match (50%), experience (20%), education (15%), location and availability (10%), and text similarity between CV and job description (5%). Weights are adjustable by the administrator. Each result shows matched and missing skills so the recruiter understands why a candidate ranks highly. Personal details such as name, gender, age, photo and ethnicity are excluded from scoring. The recruiter moves candidates through the stages Applied, Screened, Shortlisted, Interview, Offer, and Hired or Rejected; interviews are scheduled and notifications sent by email and SMS. Managers see dashboards on time to shortlist, time to hire and vacancy status, and every important action is recorded in an audit log.

2.3 Deliverables

Deliverable	Description / Measurable details
Deployed web application	8 modules live on a public HTTPS URL, 4 user roles, 9 MongoDB collections (users, employers, vacancies, candidateProfiles, applications, matchResults, interviews, notifications, auditLogs).
CV upload and parsing	Accepts PDF and DOCX up to 5 MB; target of at least 85% correct extraction of skills, experience and education on a test set of 100 CVs (measured and reported).
Automatic matching engine	Score 0-100 with matched and missing skills; top-10 ranked list for 1,000 candidates in under 5 seconds; no protected personal attributes used in scoring.
Matching evaluation	Comparison with recruiters' manual shortlists on 30 vacancies using precision at top 10 and recall; target of at least 70% overlap, with results reported as measured.
Dashboard and reports	At least 6 reports (time to shortlist, time to hire, applications per vacancy, pipeline stages, vacancies by status, source of applicants); export to PDF/CSV.
Test suite	Automated unit and API tests with at least 70% coverage of server code.
Documentation and training	SRS, UML and data-flow diagrams, user manual, 3 training sessions, final report, GitHub repository with setup instructions, defense presentation.

3. Feasibility Assessment

3.1 Economic Feasibility: Project Costs and Benefits

Cost item	Basis	Cost (Ksh)
Project manager (part-time)	16 weeks	160,000
Systems analyst	4 months at 60,000	240,000
2 full-stack MERN developers	4 months at 80,000 each	640,000
Data/ML engineer (matching service)	3 months at 70,000	210,000
UI/UX designer	6 weeks	90,000
QA tester	3 months at 45,000	135,000
Development devices and test equipment	Shared, one-off	100,000

Cloud hosting, MongoDB, CV file storage, domain and SSL	First year	80,000
Email and SMS gateway	First year	40,000
AI/NLP API credits	Development and first year	70,000
User training and manuals	3 sessions	50,000
Data protection compliance review	One-off, legal advice and registration	80,000
Contingency	10% of Ksh 1,895,000	189,500
TOTAL PROJECT BUDGET		2,084,500

Estimated benefits (assumptions to be validated during analysis): about Ksh 600,000 per year in recruiter time saved through automatic CV parsing and ranking; about Ksh 700,000 per year in extra placement revenue from faster shortlisting and reuse of the old candidate database; about Ksh 150,000 per year in reduced advertising and administration. Total about Ksh 1,450,000 per year. Annual running cost (hosting, gateways, AI API, maintenance) is about Ksh 200,000, giving net benefit of about Ksh 1,250,000 per year. Payback period is about 1.7 years (20 months); 5-year net present value at 12% is about Ksh 2.42 million, so the project is economically feasible. Intangible benefits include a better candidate experience, consistent records and fairer, explainable screening.

3.2 Technical Difficulties and Mitigation

Difficulty / risk	Mitigation
CVs have many layouts; scanned or image-only CVs cannot be read	Support text-based PDF and DOCX first; applicant can correct the extracted profile; image CVs are flagged for manual review.
The same skill is written differently (for example JS and JavaScript)	Maintain a skills dictionary with synonyms, edited by administrators and updated from recruiter feedback.
Bias or unfair ranking	Exclude name, gender, age, photo and ethnicity from scoring; show explanations; recruiter makes every final decision; review of results by group in testing.
Few labeled examples to test matching quality	Recruiters manually shortlist 30 past vacancies to form the test set.
Performance with thousands of CVs; AI service outage or cost	Background processing, database indexes, caching; fall back to keyword scoring if the AI service is unavailable.
Candidate data is sensitive	Role-based access, HTTPS, encrypted storage, hashed passwords, audit log, external review before go-live.

3.3 Operational, Schedule, Legal and Contractual, and Political Feasibility

- **Operational:** recruiters already follow the same stages manually, so the system fits current work. Three training sessions and a manual support adoption. Feasible.
- **Schedule:** 16 weeks with a 6-person team in eight 2-week sprints, with testing and buffer in Weeks 13 to 16. Feasible if scope changes are controlled.
- **Legal and contractual:** compliance with the Kenya Data Protection Act, 2019 (registration as data controller, consent when CVs are collected, data minimization, retention limits, right to correct or delete a profile, processing terms with hosting and AI providers); equal-opportunity rules in hiring; contract clauses on software ownership, acceptance criteria, payment milestones, confidentiality and support after go-live.
- **Political:** the Managing Director supports the project. Recruiters may fear replacement, so the system is presented as an assistant and recruiters keep all decisions. Employers must agree to post vacancies through the portal, which the Head of Recruitment will promote.

3.4 High-Level Project Schedule and Major Milestones

Milestone	Expected completion	Description / Details
M1: Plan approved	Fri 22 Jan 2027 (Weeks 1-2)	BPP signed by sponsor; stakeholder interviews and literature review done.
M2: Requirements and design signed off	Fri 5 Feb 2027 (Weeks 3-4)	SRS, UML and data-flow diagrams, database schema, wireframes and API specification approved.
M3: Foundation ready	Fri 19 Feb 2027 (Weeks 5-6)	Project structure, GitHub workflow, login with 4 roles working on a test server.

M4: Vacancies, profiles and applications	Fri 12 Mar 2027 (Weeks 7-9)	Vacancy management, CV upload and parsing, application pipeline, data import.
M5: Matching and dashboard	Fri 2 Apr 2027 (Weeks 10-12)	Matching engine with explanations, interviews, notifications, dashboard.
M6: Testing and evaluation	Fri 16 Apr 2027 (Weeks 13-14)	All tests done; matching evaluated against manual shortlists; defects fixed.
M7: Go-live and handover	Fri 30 Apr 2027 (Weeks 15-16)	Deployment, training, user manual, final report and presentation.

4. Management Issues

4.1 Management Concerns

- **Fairness:** management must be able to show that candidates are not discriminated against (explained scores, excluded personal attributes, human decisions).
- **Privacy** of candidate CVs and compliance with the Data Protection Act (access control, consent, retention rules).
- **Budget overruns and scope creep** (10% contingency; change control; sponsor approval for changes above 5% of budget).
- **Disruption during rollout** (staged go-live with a 2-week parallel run beside Excel), employer and staff adoption, and dependence on a few developers (documentation and code reviews).

4.2 Team Composition

Role	**Responsibility**
Sponsor (Managing Director) and Head of Recruitment	Approve budget and milestones; provide data and users; accept deliverables.
Project manager	Schedule, risks, budget tracking and status reports.
Role	**Responsibility**
Systems analyst: Victorious Chepchirchir Korir	Requirements, feasibility, UML modelling, BPP, acceptance testing coordination.
2 full-stack developers and data/ML engineer	React front end, Express API, MongoDB; CV parsing and matching service.
UI/UX designer and QA tester	Wireframes and usability; test plans, functional and security testing.

4.3 Communication Plan

Communication	**Frequency**	**Audience and content**
Written status report (1 page)	Weekly, every Friday	Sponsor, Head of Recruitment: progress against milestones, tasks done, budget spent, risks and issues, plan for next week.
Sprint review and demo	Every 2 weeks	Recruiters and stakeholders: working increment, feedback and backlog changes.
Milestone report and sign-off	At each milestone (M1-M7)	Sponsor: deliverables completed, acceptance decision.
Steering meeting	Monthly	Sponsor and management: budget, schedule, change requests.
Daily stand-up (15 minutes)	Daily	Project team: yesterday, today and blockers.

4.4 Project Standards and Procedures

- **Modelling and documentation:** UML 2.x and data-flow diagrams; REST API documented with OpenAPI; README and setup instructions in GitHub.
- **Coding and version control:** GitHub branching with pull-request review before merging; ESLint and Prettier; clear commit messages; no secrets in the repository.
- **Quality and security:** automated tests (Jest/Vitest and Supertest) with at least 70% coverage; OWASP API Security Top 10 checklist; fairness review of matching results before go-live.
- **Change control and reporting:** every change request is logged, assessed for cost and schedule, and approved by the sponsor; this BPP is updated whenever scope, budget or schedule changes.